

Chapter 6: Food borne Diseases

Intended Learning Objectives

By the end of this chapter, students should be able to:

1. Define foodborne diseases and distinguish between infection and intoxication.
 2. Classify foodborne diseases based on causative agents and mechanisms.
 3. Describe the clinical features and transmission routes of key pathogens including Cholera, Poliomyelitis, Hepatitis A, Typhoid and Brucellosis.
 4. Recognize the role of food handlers and environmental factors in disease spread.
 5. Recognize preventive measures including vaccination to reduce the risk of foodborne outbreaks.
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1. Overview of Foodborne Diseases

Foodborne diseases result from ingestion of food contaminated with infectious agents or their toxins. These may be classified as:

- **Food Infection:** Caused by ingestion of live pathogens that multiply in the host.
- **Food Intoxication:** Caused by ingestion of preformed toxins in food.

They represent a major public health concern, especially in developing countries where food safety systems may be under-resourced.

2. Classification of Foodborne Diseases

Category	Examples
Bacterial Infections	<i>Salmonella, Shigella, E. coli, Listeria, Brucella, Vibrio cholerae</i>
Bacterial Intoxications	<i>Staphylococcus aureus, Clostridium botulinum, Bacillus cereus</i>
Viral Infections	Hepatitis A virus, Norovirus, Poliovirus
Parasitic Infections	<i>Giardia lamblia, Entamoeba histolytica, Taenia saginata</i>
Chemical Poisoning	Pesticides, heavy metals, food additives
Fungal Toxins	Aflatoxins from <i>Aspergillus</i> species

3. Selected food borne diseases

3.1. Cholera

- **Agent:** *Vibrio cholerae*, a gram-negative bacterium
- **Transmission:** Fecal-oral route via contaminated water or food, especially raw seafood and vegetables irrigated with contaminated water
- **Incubation Period:** 2 hours to 5 days
- **Clinical Features:** Profuse watery diarrhea (“rice-water stools”), vomiting, rapid dehydration, hypotension, shock

- **Complications:** Death due to severe dehydration if untreated
- **Prevention:**
 - Safe drinking water and sanitation
 - Proper food handling and cooking
 - Health education in endemic areas
 - Surveillance and outbreak response

Vaccination Details

- **Type:** Oral cholera vaccine (e.g., Dukoral®, Shanchol®)
 - **Route:** Oral
 - **Dose:**
 - Adults and children ≥ 2 years: 2 doses, 1–6 weeks apart
 - **Protection Duration:** Up to 2 years
 - **Indications:**
 - Travelers to endemic areas
 - Populations in outbreak zones
 - Refugee camps and disaster settings
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3.2. Poliomyelitis (Polio)

- **Agent:** Poliovirus (types 1, 2, and 3), an enterovirus
- **Transmission:** Fecal-oral route via contaminated food, water, or hands

- **Reservoir:** Humans
- **Incubation Period:** 7–14 days
- **Clinical Features:**
 - Most infections are asymptomatic
 - Minor illness: fever, sore throat, headache
 - Major illness: aseptic meningitis, flaccid paralysis, respiratory failure
- **Complications:** Permanent paralysis, post-polio syndrome
- **Prevention:**
 - Hygiene and sanitation
 - Mass immunization campaigns

Vaccination Details

- **Types:**
 - **Oral Polio Vaccine (OPV)** – live attenuated
 - **Inactivated Polio Vaccine (IPV)** – injectable
- **Route:**
 - OPV: Oral drops
 - IPV: Intramuscular or subcutaneous
- **Schedule** (Egypt national program):
 - OPV at birth, 2, 4, and 6 months
 - IPV at 4 and 6 months

- Booster doses at 18 months and 4 years
 - **Indications:**
 - All children under 5
 - Travelers to polio-endemic regions
 - Healthcare workers in outbreak zones
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3.3. Hepatitis A

Causative Agent:

Hepatitis A virus (HAV), a non-enveloped RNA virus from the Picornaviridae family.

Reservoir:

Humans (no chronic carrier state).

Mode of Transmission:

Fecal-oral route via contaminated food, water, or close personal contact.

Incubation Period:

15–50 days (average 28 days)

Clinical Features:

- Fever, malaise, nausea, abdominal discomfort
- Jaundice, dark urine, pale stools
- Often asymptomatic in children

Diagnosis:

- Detection of anti-HAV IgM antibodies
- Elevated liver enzymes (ALT, AST)

Prevention and Control:

- Hepatitis A vaccine
- Improved sanitation and hygiene
- Safe food and water practices
- Post-exposure prophylaxis with vaccine or immunoglobulin

Epidemiological Notes:

- Endemic in areas with poor sanitation
- Outbreaks common in schools, food service, and refugee settings
- Surveillance essential for outbreak detection

3.4. Typhoid Fever

Causative Agent:

Salmonella enterica serotype Typhi (Gram-negative bacillus)

Reservoir:

Humans (chronic carriers may shed bacteria for years)

Mode of Transmission:

Fecal-oral route via contaminated food or water

Incubation Period:

7–14 days

Clinical Features:

- Sustained high fever, abdominal pain
- Rose-colored spots on trunk
- Constipation or diarrhea, hepatosplenomegaly

Diagnosis:

- Blood culture (gold standard in early phase)
- Stool and urine cultures
- Widal test (limited specificity)

Prevention and Control:

- Typhoid vaccination (Vi polysaccharide or conjugate vaccine)
- Safe water and food hygiene
- Identification and treatment of carriers
- Public health education

Epidemiological Notes:

- Endemic in many low- and middle-income countries
- Risk increases during floods and poor sanitation
- WHO recommends vaccination in high-risk populations

3.5. Brucellosis

Causative Agent:

Brucella species (e.g., *B. melitensis*, *B. abortus*, *B. suis*)

Reservoir:

Domestic animals (goats, sheep, cattle, pigs)

Mode of Transmission:

- Direct contact with infected animals or products
- Ingestion of unpasteurized dairy
- Inhalation in laboratory settings

Incubation Period:

5 days to several weeks

Clinical Features:

- Undulating fever, night sweats
- Arthralgia, fatigue, weight loss
- Hepatosplenomegaly, lymphadenopathy

Diagnosis:

- Blood culture (slow-growing organism)
- Serological tests (e.g., ELISA, SAT)
- PCR in specialized labs

Prevention and Control:

- Pasteurization of milk
- Animal vaccination and control

- Protective measures for high-risk occupations
- Notification and surveillance

Epidemiological Notes:

- Zoonotic disease with occupational relevance
 - Endemic in Mediterranean, Middle East, Latin America
 - Often underreported due to nonspecific symptoms
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4. General Sources of Contamination

- Infected food handlers
 - Improper cooking or storage
 - Cross-contamination between raw and cooked food
 - Contaminated water used in food preparation
 - Poor sanitation and hygiene practices
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5. Prevention and Control

Strategy	Examples
Food Safety Practices	Handwashing, clean utensils, safe cooking temperatures
Public Health Measures	Surveillance, outbreak investigation, food inspection

Strategy	Examples
Education and Awareness	Community health campaigns, school programs
Vaccination	Hepatitis A, Typhoid, Cholera, and Polio vaccines for high-risk groups
Regulation and Enforcement	Food labeling, hygiene standards, penalties for violations

End-of-Chapter Assessment

Multiple Choice and True/False Questions

1. True or False: Cholera is primarily transmitted through airborne droplets.

Answer: False

2. Which vaccine is administered orally to prevent polio?

- A. IPV
- B. OPV
- C. TCV
- D. Hepatitis A vaccine

Answer: B

3. **Hepatitis A is commonly transmitted through contaminated blood and body fluids.**

False

4. Typhoid fever can be spread by chronic human carriers who shed *Salmonella Typhi* in their stool.

True

5. Brucellosis is a viral infection primarily spread through airborne droplets in crowded urban settings.

False